
Algebraic Stacks
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These are currently incomplete lecture notes for a course on algebraic stacks given in the 2021 Winter term at the University of Toronto. They include the notetakers own marginalia throughout, included without warning into the material presented in lecture, which itself has been rearranged. Any mistakes in these notes are entirely those of the notetaker.

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1 Grothendieck topologies

1.1 Grothendieck's philosophy of points

Let C be any category and $X \in \text{Obj}(C)$. Then there is a functor $h_X: {}^{\text{op}}\mathbf{Set} \rightarrow \mathbf{Set}$ defined by $h_X(Y) = \text{Hom}_C(Y, X)$. The main point is that h_X determines X uniquely.

Lemma 1 (Yoneda). *Let $F: C^{\text{op}} \rightarrow \mathbf{Set}$ be a functor. Then there is a natural bijection*

$$\text{Nat}(h_X, F) \simeq F(X).$$

Proof. Exercise. Note that showing naturality in detail requires more space than most expositions assign to this proof. $\square$

Corollary 1. Let $X_1, X_2 \in \text{Obj}(C)$. Then

$$\text{Nat}(h_{X_1}, h_{X_2}) \simeq \text{Hom}_C(X_1, X_2).$$

That is to say, C is a full subcategory of $\mathbf{Set}^{C^{\text{op}}}$.

Let C be the category of schemes (or schemes over a base scheme S). Which contravariant functors $C \rightarrow \mathbf{Set}$ are representable?

What about automorphisms?

Definition 1. Let k be a field and X a scheme. Recall that a k -point of X is an element of $\text{Hom}_{\mathbf{Sch}}(\text{Spec } k, X)$. Likewise if S is any other scheme, an S point of X is an element of $\text{Hom}_{\mathbf{Sch}}(S, X)$.

So far we have dealt with functors $\mathbf{Sch}^{\text{op}} \rightarrow \mathbf{Set}$.

Definition 2. A *groupoid* is a category where every morphism is an isomorphism. We write $\mathbf{Gpd}$ for the category of groupoids.

Example 1. A set is a discrete groupoid.

If C is a groupoid and $\bar{C}$ is the set of isomorphism classes of $\text{Obj}(C)$, then we can picture $\bar{C}$ is

picture

where the G_i are automorphism groups.

1.2 Grothendieck topologies

The idea behind stacks. We want to study functors $\mathbf{Sch}^{\text{op}} \rightarrow \mathbf{Gpd}$. These will define points of a stack. To keep things manageable, we must also impose conditions on these functors. In order to state these conditions, we will need the notion of a Grothendieck topology.

Example 2. Let X be a topological space and $C_X = \text{Open}(X)$ the category of open sets of X . Given a collection of open subsets $U_i \hookrightarrow U$, there is a notion that $\{U_i\}_i$ covers U . Recall that $\text{Presh}(X) := \mathbf{Set}^{C_X^{\text{op}}}$. To say that a presheaf is a sheaf, we need the notion of covering. But we also need *only* the notion of covering.

Definition 3. Let C be a category with all fibre products. A *Grothendieck topology* on C is a class of coverings $\{U_i \rightarrow U\}_{i \in I}$ that is we choose for each $U \in \text{Obj}(C)$ which collections of morphisms to U are to be called coverings, such that

1. Coverings are preserved by base-change: if $\{U_i \rightarrow U\}_{i \in I}$ covers U then if $V \rightarrow U$ $\{U_i \times_U V \rightarrow V\}_{i \in I}$ covers V .
2. If $\{U_i \rightarrow U\}_{i \in I}$ is a covering, and if $\{V_{ij} \rightarrow U_i\}_{j \in J_i}$ is a covering of each U_i , then $\{V_{ij} \rightarrow U_i \rightarrow U\}_{(i,j) \in I \times J}$ is a cover of U .
3. If $U \rightarrow V$ is isomorphism, then it is a covering of V .

Definition 4. A category C with a Grothendieck topology on it is called a *site*.

Example 3. If X is a topological space, then $C = \text{Open}(X)$ with morphisms given by inclusions, is a Grothendieck topology.

Remark 1. If $C = \text{Open}(X)$ then fibre products are intersections, and the first condition says that if $U = \bigcup_i U_i$ and $V \hookrightarrow U$, then $V = \bigcup_i (U_i \cap V)$ and $U_i \cap V$ is open in V .

Example 4. If $C = \mathbf{Top}$, setting $\text{Cov}(X) = \{(X_i \rightarrow X)_i \mid X_i \subset X \text{ is open } \bigcup_i X_i \twoheadrightarrow X\}$.

There is a general formalism for constructing sites out of schemes (or other geometric objects).

Definition 5. Let P and Q be two properties of morphisms. A P - Q -site on a scheme Y is the full subcategory of $\mathbf{Sch}/Y$ whose objects are morphisms $X \rightarrow Y$ which have property P and coverings of such a morphism $X \rightarrow Y$ are collections of morphisms

$$\begin{array}{ccc} X_i & \xrightarrow{\quad} & X \\ & \searrow \quad \swarrow & \\ & Y & \end{array}$$

in $\mathbf{Sch}/Y$ such that $X_i \rightarrow X$ has property Q and $\coprod_i X_i \rightarrow X$.

Remark 2. If Y is not specified, take $Y = \text{Spec } \mathbb{Z}$.

Definition 6. If P is vacuous, we get the *big P site*.

Definition 7. When $P = Q$ we get the *small Q site*. (When indeed P and Q are such that we get a site.)

Example 5. The big and small Zariski site (with Q the property of being an open embedding) is a site. Similarly, we can work with the big or small étale sites. We will also see examples to do with the properties of being smooth or flat, among others.

1.3 Sheaves

Given a site C one can define the notion of a sheaf of sets on C . A *presheaf* on C is just any functor $C^{\text{op}} \rightarrow \mathbf{Set}$.

Definition 8. A presheaf $F: C^{\text{op}} \rightarrow \mathbf{Set}$ is a *sheaf on the site C* if for all coverings $\{U_i \rightarrow U\}_{i \in I}$ the following diagram of sets is an equalizer diagram

$$F(U) \longrightarrow \prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \times_U U_j).$$

Here the two parallel arrows correspond to the arrows $U_i \times_U U_j \rightarrow U_i$ and $U_i \times_U U_j \rightarrow U_j$.

Example 6. Let $C = \mathbf{Sch}$ with the Zariski topology: Morphisms are morphisms of schemes, and a covering means that $\bigcup_i U_i \rightarrow X$ is surjective (on the level of topological spaces) with $U_i \hookrightarrow X$ open immersions. A functor $\mathbf{Sch}^{\text{op}} \rightarrow \mathbf{Set}$ is just a presheaf on this site.

Equivalently, we can work with $\mathbf{Sch}_{\text{open}}$, the subcategory where we only consider open immersions of schemes. We can define the Zariski topology on it the same way.

Definition 9. Let F, G be two presheaves on C , where C is any site. A natural transformation $\alpha: F \rightarrow G$ is *representable* if for all $h_X \rightarrow G$, $X \in \text{Obj}(C)$, then

$$h_X \times_G F: C^{\text{op}} \rightarrow \mathbf{Set}$$

is representable by an object $Y \in \text{Obj}(C)$.

The main point is that the “fibres” of α all belong to C .

We can define the notion of covering also for presheaves.

Definition 10. A collection $\{F_i \rightarrow F\}_{i \in I}$ is an *open cover* if

1. all $F_i \rightarrow F$ are representable.
2. for all $h_X \rightarrow F$ such that $h_X \times_F F_i \simeq h_{Y_i}$, $\{Y_i \rightarrow X\}_{i \in I}$ is a cover in C .

Lemma 2. A functor $F: \mathbf{Sch}^{\text{op}} \rightarrow \mathbf{Set}$ is representable iff

1. F is a sheaf in the Zariski topology.
2. There is a covering $\{F_i \rightarrow F\}_{i \in I}$, where every F_i is representable.

That is, “something is a scheme if it satisfies the sheaf condition and is locally a scheme.”

1.4 Examples of representable functors for schemes

Fix some base scheme S (i.e. $\text{Spec } k$ for a field k). Often it is convenient to at least assume S is Noetherian. We have discussed functors $h_X: \mathbf{Sch}^{\text{op}} \rightarrow \mathbf{Set}$ and how they determine the S -scheme X entirely.

1. $X = \mathbb{A}_S^n$ (assume $S = \text{Spec } A$ for simplicity). Then

$$h_X(Y) = \text{Hom}_{\mathbf{Sch}}(Y, \mathbb{A}^n) = \Gamma(\mathcal{O}_Y)^n.$$

More generally, if $X \subset \mathbb{A}^n$ is a closed subscheme, then let $I \subset R[x_1, \dots, x_n]$ be the ideal defining X . Then

$$h_X(Y) = \{(f_1, \dots, f_n) \mid f_i \in \mathcal{O}_Y(Y), g(f_1, \dots, f_n) = 0 \forall g \in I\}.$$

2. $X = \mathbb{A}^n \setminus \{(0, \dots, 0)\}$. Then

$$h_X(Y) = \{(f_1, \dots, f_n) \mid f_i \in \mathcal{O}_Y(Y)^{n+1}, (f_1, \dots, f_n) = \mathcal{O}_Y(Y)\}.$$

That is, we require for all $\mathfrak{p}$ $(f_i)_{\mathfrak{p}} \notin \mathfrak{m}_{\mathfrak{p}}$ for some i .

3. $X = \mathbb{P}^n = (\mathbb{A}^{n+1} \setminus \{(0, \dots, 0)\})/\mathbb{G}_m$. A first attempt would be to try to define

$$h'_X(Y) = \{(f_1, \dots, f_n) \in \mathcal{O}_Y(Y)^{n+1} \mid (f_i)_i = (1)\} / \mathcal{O}_Y(Y)^{\times}.$$

Elements of the RHS do give morphisms to the projective space, but not all: some morphisms to projective space do not lift to morphisms to affine space without the origin. For example, $\text{id}: \mathbb{P}^n \rightarrow \mathbb{P}^n$. There are no nonconstant global functions for us to use!

The problem is that h'_X is not a sheaf in the Zariski topology, we need to sheafify it. The point is that *locally* all maps to $\mathbb{P}^n$ can be lifted.

1.4.1 Hilbert schemes

We want a scheme that “parameterizes” subschemes of a given scheme X e.g. $\mathbb{P}_k^n$. We want to look at schemes with given Hilbert polynomial.

Let $Z \subset \mathbb{P}^n$. Then $\mathcal{O}_Z$ is a coherent sheaf. Set $\tilde{f}(i) = \dim \Gamma(\mathcal{O}_Z \otimes \mathcal{O}_{\mathbb{P}^n}(i))$. If $i \gg 0$, $\tilde{f}(i)$ is a polynomial $f(i)$.

Fix a polynomial f . We want $\text{Hilb}_f(\mathbb{P}_k^n)$ to be a scheme which parameterizes closed subschemes of $\mathbb{P}_k^n$ with Hilbert polynomial f . We want a functor $F: \mathbf{Sch}_k^{\text{op}} \rightarrow \mathbf{Set}$ such that

$$F(Y) = \text{Hom}(Y, \text{Hilb}_f(\mathbb{P}_k^n)).$$

Example 7. Let $m > 0$. Then $\text{Hilb}_m(\mathbb{P}_k^n)$ is the *Hilbert scheme of m points* with $f(i) = m$ i.e. zero dimensional subschemes of length m : $Z \subset \mathbb{P}_k^n$ such that $\dim Z = 0$ and $\dim \mathcal{O}_Z(Z) = m$. For example, Z could be m disjoint closed points. We define

$$F(Y) = \{Z \subset Y \times \mathbb{P}^n \mid Z \text{ closed, flat over } Y, \text{ the fibre of } Z \text{ over every point of } Y \text{ has Hilbert polynomial } f\}$$

Definition 11. Recall that $Z \rightarrow Y$ is *flat* if $\mathcal{O}_Z$ is flat over $\mathcal{O}_Y$. For affine schemes, this just means that one ring is flat over the other.

Theorem 1. *The functor $F = F_f$ is representable by a projective scheme over k .*

Exercise. Think what to do if the base scheme S has subschemes itself.

If $X \subset \mathbb{P}_k^n$ is a any closed subscheme, then we can talk about $\text{Hilb}_f(X) \subset \text{Hilb}_f(\mathbb{P}^n)$.

If $f = m$ is constant and $U \subset \mathbb{P}^n$ is open, then $\text{Hilb}_f(U)$ is just an open subset of $\text{Hilb}_f(\mathbb{P}^n)$. Beyond zero dimensional subschemes, this is not necessarily true.

The Hilbert scheme of points is well-defined for any quasiprojective X . Flatness plus connectedness of Y implies the Hilbert polynomial is constant on fibres.

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Theorem 2 (Hartshorne). *The scheme $\text{Hilb}_f(\mathbb{P}^n)$ is connected (but can be very reducible).*

Remark 3. This can be proven without recourse to representability.

Exercise 1. *Given a scheme X , define what it means for X to be connected just in terms of the functor h_X .*

Proof. Recall that a scheme X is connected iff $\text{Spec } \mathcal{O}_X(X)$ is connected, which happens iff $\mathcal{O}_X(X)$ has no idempotents other than 1. Hence if X is affine, and notice that idempotents of A are given by

$$\text{Hom}_{\mathbf{CRing}}(\mathbb{Z}[x]/(x^2 - x), A) \simeq \text{Hom}_{\mathbf{Sch}}(\text{Spec } A, \text{Spec } \mathbb{Z}[x]/(x^2 - x)) \simeq \text{Hom}_{\mathbf{Sch}^{\text{op}}}(h_{\text{Spec } A}, h_{\text{Spec } \mathbb{Z}[x]/(x^2 - x)}).$$

The scheme $\text{Spec } A$ is connected iff these sets are all singletons. Thus in general, we require that

$$\text{Hom}_{\mathbf{Sch}^{\text{op}}}(h_X, h_{\text{Spec } \mathbb{Z}[x]/(x^2 - x)})$$

be a singleton. □

1.4.2 Quot schemes

Let $\mathcal{F}$ be any coherent sheaf on $\mathbb{P}_k^n$. Fix a polynomial f . Define $\text{Quot}_f(\mathbb{P}_k^n)$ to “parameterize” quotients $\mathcal{G}$ of $\mathcal{F}$ with Hilbert polynomial f . Define a functor by

$$F_f(Y) = \{\mathcal{F} \boxtimes \mathcal{O}_Y \rightarrow \mathcal{G} \mid \mathcal{G} \in \text{Coh}(Y \times \mathbb{P}^n) \text{ } \mathcal{G} \text{ is flat over } Y, \forall y \in Y, \mathcal{G}|_y \text{ has Hilbert polynomial } f\},$$

where by $\mathcal{G}|_y$ we mean $\mathcal{G}$ restricted to the fibre over y .

Theorem 3 (Grothendieck). *The functor F_f is representable by a projective scheme over k .*

1.4.3 More explicit examples of Hilbert schemes

Consider $\text{Hilb}_m(k^2)$ the Hilbert scheme of m points in the plane $\mathbb{A}^2$. Suppose that $k = \bar{k}$. A closed subscheme of $\mathbb{A}_k^2$ is the same as an ideal $I \subset k[x, y]$ such that $\dim_k k[x, y]/I = m$.

Theorem 4. $\text{Hilb}_m(\mathbb{A}_k^2)$ is a smooth connected scheme of dimension $2n$.

Suppose X is a scheme and $x \in X(k)$. Then recall that $T_x X$ is in bijection with maps α such that

$$\begin{array}{ccc} \text{Spec } k[\epsilon]/(\epsilon^2) & \xrightarrow{\alpha} & X \\ \uparrow & & \uparrow \\ \text{Spec } k & \longrightarrow & \{x\} \end{array}$$

commutes.

Definition 12. For any functor $F: \mathbf{Sch}_k^{\text{op}} \rightarrow \mathbf{Set}$, a k -point of F is an element of $F(\text{Spec } k)$. For all $x \in F(\text{Spec } k)$, we define

$$T_x F = \{\alpha \in F(\text{Spec } k[\epsilon]/(\epsilon^2)) \mid \alpha|_{\text{Spec } k} = x\},$$

that is $\alpha \mapsto x$ under the function $F(\text{Spec } k[\epsilon]/(\epsilon^2)) \rightarrow F(\text{Spec } k)$ induced by the morphism of schemes induced by the ring map sending $\epsilon \mapsto 0$.

How can we describe the tangent space at a point of $I \in \text{Hilb}_m(\mathbb{A}_k^2)$?

Proposition 1. *We have $T_I \text{Hilb}_m(\mathbb{A}_k^2) \simeq \text{Ext}_{k[x, y]}^1(I, I)$.*

Proof. We will define maps in both directions.

By definition, an element of $T_I \text{Hilb}_m(\mathbb{A}^2)$ is an ideal $\tilde{I} \subset (k[\epsilon]/\epsilon^2)[x, y]$ flat over $k[x, y]$ such that $\tilde{I}/\epsilon = I$. Flatness over the dual numbers is equivalent to freeness over the dual numbers, and so we obtain a short exact sequence of $k[\epsilon]/\epsilon^2$ -modules

$$0 \longrightarrow \epsilon \tilde{I} \longrightarrow \tilde{I} \longrightarrow I \longrightarrow 0.$$

Fixed is therefore an isomorphism $\varphi: \tilde{I}/\epsilon \tilde{I} \rightarrow I$. Consider the map $\cdot \epsilon: \tilde{I} \rightarrow \tilde{I}$ killing $\epsilon \tilde{I}$. Thus we get a map $\tilde{I}/\epsilon \tilde{I} \rightarrow \tilde{I}$. Therefore we obtain a composite map η given by

$$I \xrightarrow{\varphi^{-1}} \tilde{I}/\epsilon \tilde{I} \longrightarrow \tilde{I}$$

by definition with image $\epsilon \tilde{I}$. This gives a self-extension

$$0 \longrightarrow I \xrightarrow{\eta} \tilde{I} \longrightarrow I \longrightarrow 0.$$

In the other direction, given an extension M of I by itself, we can make M into a $(k[\epsilon]/\epsilon^2)[x, y]$ -module by saying how ϵ acts. Let ϵ kill $I \subset M$ and have image I . That is, we define the action of ϵ by requiring that

$$\begin{array}{ccc} M/I & \xrightarrow{\epsilon} & I \\ \downarrow \sim & \nearrow \text{id} & \\ I & & \end{array}$$

commutes.

Exercise 2. Construct the embedding of M into $(k[\epsilon]/\epsilon^2)[x, y]$ as a $(k[\epsilon]/\epsilon^2)[x, y]$ -submodule.

□

In general, Hilbert schemes can quite nasty. We give three examples of this.

For example, $\text{Hilb}_m(\mathbb{A}^3)$ is in general reducible; moreover, it has irreducible components of different dimensions. For example,

$$\text{Hilb}_m(X) \supset (X \times \underbrace{\cdots \times X}_m \setminus (\text{all diagonals}))/\mathfrak{S}_m$$

is always an open embedding. Generically, any point of the Hilbert scheme of the plane comes from this part. But for $X = \mathbb{A}^3$, this isn't true for points in the other components. Worse, if $d = \dim X$, this open part has dimension dm , so if $X = \mathbb{A}^3$, then $\text{Hilb}_m(X)$ has a component of dimension $3m$, but always has components of much higher dimensions. (See Okounkov's Park City lectures.)

Mumford studied $\text{Hilb}_f(\mathbb{P}^3)$ for $f(t) = 14t - 23$. This Hilbert scheme gives one-dimensional subschemes, smooth curves of degree 14 and genus 23. Some irreducible components of it are not reduced.

Theorem 5 (Vakil). *Every singularity of finite type over $\mathbb{Z}$ appears in some Hilbert scheme. Moreover, it is enough to consider only Hilbert schemes of surfaces in $\mathbb{P}^4$.*

2 Algebraic spaces

The general upshot from the last section is that we want to study “nice” functors from $\mathbf{Sch}^{\text{op}} \rightarrow \mathbf{Set}$. Here “nice” means several things, but at least it means that it must be a sheaf in some Grothendieck topology.

We now want, starting with schemes, to define more general “spaces” using functors of points.

2.1 The étale topology

Definition 13. Consider the category of schemes $\mathbf{Sch}$, or schemes over a based scheme S , the category $\mathbf{Sch}/S$. Let X be an object. Setting

$$\mathrm{Cov}(X) = \left\{ (V_i \xrightarrow{\varphi_i} X)_i \mid \text{the } \varphi_i \text{ are étale and jointly surjective} \right\}.$$

Defines the *étale topology* on $\mathbf{Sch}$ (or $\mathbf{Sch}/S$).

Note that taking the disjoint union, we can redefine covers to be just a single surjective étale morphism $V \rightarrow X$.

2.2 Spaces

Definition 14. A *space* X (or a space over S) is a sheaf of sets on the étale site of $\mathbf{Sch}$ (or $\mathbf{Sch}/S$).

Thus for all schemes U with get a set $X(U)$, and given $V \rightarrow U$, we get $X(U) \rightarrow X(V)$ compatible with compositions, and satisfying the sheaf axiom with respect to the étale topology.

Notice that for a functor $F: \mathbf{Sch}^{\mathrm{op}} \rightarrow \mathbf{Set}$ to be a space is a property, not a structure. This philosophy is characteristic of modern algebraic geometry, and stands in contrast to the earlier approach to schemes (and everything else), which were sets with several layers of additional structure.

Remark 4. In principal, one can replace $\mathbf{Sch}$ with the category of affine schemes when X is a sheaf; X is determined thanks to the sheaf axiom on affine opens. Formally, we have an equivalence of categories

$$\mathrm{Sh}_{\mathrm{et}}(\mathbf{Sch}) \rightarrow \mathrm{Sh}_{\mathrm{et}}(\mathbf{AffSch})$$

given by restriction.

Remark 5. An actual scheme X defines a space h_X via its functor of points:

Theorem 6. *The functor of points $U \mapsto \mathrm{Hom}_{\mathbf{Sch}}(U, X)$ is a Zariski sheaf, and an étale sheaf.*

In fact, let us also consider another topology.

Definition 15. The *fppf topology* on $\mathbf{Sch}$ or $\mathbf{Sch}/S$ is the Grothendieck topology where coverings are families $(V_i \xrightarrow{\varphi_i} U)_i$ where every φ_i is flat and locally-finitely presented, and the φ_i are jointly surjective.

Definition 16. Recall that a morphism of rings $A \rightarrow B$ is *finitely-presented* if B is isomorphic to $A[x_1, \dots, x_n]/I$ as an A -module, where I is finitely-generated.

Definition 17. A morphism of schemes is *faithfully-flat* if it is flat and surjective.

Theorem 7 (Faithfully-flat descent). *If X is a scheme, then $U \mapsto X(U) = \mathrm{Hom}_{\mathbf{Sch}}(U, X)$ is an fppf-sheaf.*

Here *descent* is meant in the sense that if φ is faithfully flat and f is any morphism, there is a third map such that

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & U \\ & \searrow f & \nearrow \text{---} \\ & X & \end{array}$$

That is, f can “descend” to U when the two maps $V \times_U V \rightrightarrows X$ coincide. In terms of maps of sets, this condition means that f is constant on the fibres of φ .

Exercise 3. *Give an example of a surjective morphism $V \twoheadrightarrow U$ of schemes a map $f: V \rightarrow X$ satisfying this condition, but which does not descend. Thus flatness is necessary.*

2.3 Algebraic spaces

Definition 18. Let X, Y be spaces. We say that $\varphi: X \rightarrow Y$ is *schematic* if for all $f: Z \rightarrow Y$, where Z is a scheme, the fibre product $X \times_Y Z$ is also a scheme.

Informally, this says that the fibres of φ are schemes:

$$\begin{array}{ccc} X \times_Y Z & \longrightarrow & X \\ \downarrow & & \downarrow \varphi \\ Z & \longrightarrow & Y. \end{array}$$

If $f: X \rightarrow Y$ is a schematic morphism of spaces, then we can say that it has property P of morphisms of schemes if for all all schemes $Z \rightarrow Y$, the morphism $X \times_Z Y \rightarrow Z$ of schemes has P .

With that said, we can state the following

Definition 19. A space X is *algebraic* if

1. $\Delta: X \rightarrow X \times X$ is schematic and quasicompact
2. There is a morphism $f: X' \rightarrow X$ where X' is a scheme, where f is schematic, etale, and surjective.

That is, modulo the first technical condition, a space is algebraic if it is etale-locally a scheme.

Example 8. Let X' be a quasiseparated scheme, and let G be a finite group acting freely on X' . Then one can define X'/G , which in general will be an algebraic space. Define $X := X'/G$ by defining $X(U)$ to classify G -torsors in the etale topology $U' \rightarrow U$, together with a G -equivariant map $U' \rightarrow X'$.

We will first consider the baby case, with sets instead of schemes. Let G act on X' freely. Then $X = X'/G$ of course exists as a set. Say U is another set. We want to describe a map $U \rightarrow X$ in terms of X' .

$$\begin{array}{ccc} U' = U \times_X X' & \xrightarrow{\alpha} & X' \\ & & \downarrow \\ U & \longrightarrow & X. \end{array}$$

The set U' has a free G -action given by

$$g \cdot (u, x') = (u, g \cdot x')$$

and α is G -equivariant.

Definition 20. A G -torsor over U is a set U' with a free G -action such that $U = U'/G$.

Hence maps $U \rightarrow X$ are the same as G -torsors $U' \rightarrow U$ with a G -equivariant morphism $\alpha: U' \rightarrow X'$; a map from the quotient U'/G is the same as a map from U' constant on G -orbits.

Definition 21. A morphism $U' \rightarrow U$ is a G -torsor in the etale topology if U' is a scheme with G -action such that that locally in the etale topology U' is isomorphic to $U \times G$.

What this means explicitly is that there is an etale cover $V \rightarrow U$ such that

$$U' \times_U V \simeq (U \times G)_U V$$

are isomorphic as functors.

We have defined the space $X = X'/G$ for X' a scheme. This space is algebraic, and here finiteness of G is important. The etale surjective cover by a scheme is precisely $X' \rightarrow X$, the “quotient map:”

$$\begin{array}{ccc} U' & \longrightarrow & X' \\ \downarrow & & \downarrow \\ U & \longrightarrow & X. \end{array}$$

By definition (recall that $X \rightarrow X'$ is not a morphism of schemes, just a schematic morphism) this means that $X' \rightarrow X$ is etale and surjective.

we have to check the first property now. We have

$$\begin{array}{ccc} G \times X' & \xrightarrow{\alpha} & X' \times X' \\ \downarrow & & \downarrow \\ X & \xrightarrow{\Delta} & X \times X \end{array}$$

where $\alpha(g, x) = (x, g \cdot x)$. We need to say that α is quasiseparated if X' is quasiseparated. This is obvious, however: Any morphism from a quasiseparated scheme is quasiseparated.

Formally, according to the definition of schematic morphism, we should check property 1 for all morphisms of schemes to $X \times X$. However, quasiseparatedness is an etale-local property, so it suffices to check it on this one cover. In a precise sense, the example of X'/G is characteristic of algebraic spaces.

Proposition 2. *Any algebraic space is a quotient of a scheme by an etale equivalence relation, where δ is quasicompact.*

We'll now define the terms in the proposition.

2.3.1 Equivalence relations

Definition 22. If X_0, X_1 are schemes, an *equivalence relation* is a monic morphism $\delta: X_1 \rightarrow X_0 \times X_0$. This copies the notion of an equivalence relation on a set X_0 .

Remark 6. This also works for spaces, and δ is an equivalence relation iff for all schemes U ,

$$X_1(U) \hookrightarrow X_0(U) \times X_0(U)$$

is an equivalence relation in the sense of **Set** on the set $X_0(U)$.

Definition 23. A triple (X_0, X_1, δ) is an *etale equivalence relation* if both composites

$$X_1 \longrightarrow X_0 \times X_0 \xrightarrow[p_1]{p_0} X_0.$$

are etale.

Example 9. Let G be a finite group acting freely on X_0 . Set $X_1 = G \times X_0$. We can use same map as before for $\delta: X_1 \rightarrow X_0 \times X_0$, namely $\delta(g, x) = (x, g \cdot x)$. Note G acts freely iff δ is a monomorphism.

In a precise sense, the example of X'/G is characteristic of algebraic spaces.

Proposition 3. *Any algebraic space is a quotient of a scheme by an etale equivalence relation, where δ is quasicompact.*

We still have to define word “quotient.”

Definition 24. We define the *quotient of X_0 by X_1* as the sheaf X that is the coequalizer

$$X_1 \xrightarrow[p_1 \circ \delta]{p_2 \circ \delta} X_0 \longrightarrow X.$$

(The coequalizer, like the cokernel sheaf, must be sheafified.)

Proof of proposition. Let X be an algebraic space. We must define X_0 and X_1 . Choose $X_0 \rightarrow X$ an etale cover by a scheme. Then consider the fibre product

$$\begin{array}{ccc} X_1 & \xrightarrow{\delta} & X_0 \times X_0 \\ \downarrow & & \downarrow \\ X & \xrightarrow{\Delta} & X \times X. \end{array}$$

Then δ is an étale equivalence relation the quotient by which is X .

finish.

□

Example 10. If k is a field of characteristic not two, then let $X_0 = \mathbb{A}_k^1$, and consider

$$X_1 \xrightarrow{\delta} X_0 \times_k X_0$$

where $X_1 = \Delta \coprod \{(x, -x) \mid x \in \mathbb{A}_k^1 \setminus \{0\}\}$. We claim that this is an étale equivalence relation. The quotient X will be an algebraic space but not a scheme. It is not separated, even locally. Indeed, if X were a scheme, then $\Delta: X \rightarrow X \times X$ would be a locally-closed embedding. Hence, referring to the above diagram, δ would be a locally-closed embedding by stability under base-change. However, δ in this case is not locally-closed. Hence X cannot be a scheme.

Exercise 4. Take your favourite property of morphisms of schemes, and define it for morphisms (not necessarily schematic) of algebraic spaces.

Proof. Let $f: X \rightarrow Y$ be a morphism. If Y' is an étale cover of Y , then $X \times_Y Y'$ is an algebraic space, and itself has an étale cover X' . We say that f has P if the morphism of schemes $X' \rightarrow Y'$ has P .

way to use equivalence relation fact?

□

Definition 25. If X, Y are arbitrary spaces, a morphism $X \rightarrow Y$ is *representable* if for all algebraic spaces $U \rightarrow Y$, $X \times_Y U$ is an algebraic space.

Every scheme defines an étale sheaf, and actually also defines an fppf sheaf.

Lemma 3. The same is true for algebraic spaces.

Definition 26. Let X be a space. A *quasicoherent sheaf* on X is the same as a quasicoherent sheaf $\mathcal{F}_U$ for all $U \rightarrow X$, plus some natural compatibilities to be discussed later.

Exercise 5. Think what these compatibilities should be.

3 Algebraic stacks

Here is a major example of what we want to define: the quotient of a scheme by a group action that is not necessarily free. We shall work with **Sch**/ S (in the absolute case, $S = \text{Spec } \mathbb{Z}$). Let $\mathcal{G}$ be a group scheme over S . For example, if G is a finite group, we can take $\mathcal{G} = G \times S$. This is a finite group scheme, étale over S .

Let X' be a set, and $X = X'/G$. For example if $X' = pt$, then $X = pt$, but X has forgotten about G . When the action is not free, we want to retain more than the set-theoretic quotient.

3.1 Groupoids

Recall the definition of a groupoid.

Example 11. Let X' be a set with G -action. We can define a groupoid $X = \widehat{(X', G)}$ with objects elements of X' , and for all $x, y \in X'$, setting $\text{Hom}(x, y) = \{g \in G \mid gx = y\}$.

Lemma 4. Assume G acts on X' freely. Then $X = X'/G$, and we have equivalences of categories

$$\widehat{(X', G)} \simeq \widehat{X'/G} \simeq \hat{X},$$

where $\hat{X}$ is discrete.

Proof. In one direction, a functor is given by the quotient map, on objects sending $x \mapsto [x]$. The inverse functor is not canonical, and defining it requires choosing a set of representatives for the orbits. $\square$

The lemma is false when the action is not free. In that case, the LHS groupoid should be used to define the quotient.

Example 12. Let X be a topological space, and recall the fundamental groupoid $\Pi(X)$ of X . Its objects are points of X , and morphisms are homotopy classes of paths from x to y . If X is simply-connected, we can choose any basepoint x_0 , and this category is equivalent to the one-point groupoid $\pi_1(X, x_0)$. Note that if $\mathcal{G}$ is a groupoid connected as a category, then any two objects are isomorphic. Thus X is path-connected iff $\Pi(X)$ is connected.

3.2 Stacks

We defined spaces as functors $\mathbf{Sch}^{\text{op}} \rightarrow \mathbf{Set}$ satisfying some sheaf condition. Stacks will be similar objects, but with groupoids instead of sets. So for all schemes U , we get a groupoid $X(U)$, together with restriction functors, plus the sheaf property. “Algebraic” will mean “nicely covered by a scheme.” We will first see some examples that we will eventually make precise.

Example 13. Let G be a finite group acting on a scheme X' . Want to define $X = X'/G$ such that $X(U)$ is a group. As before, we set

$$\text{Obj}(X)(U) = \{G\text{-torsors in etale top. } U' \rightarrow U, \varphi: U' \rightarrow X' \text{ } G\text{-equivariant}\}.$$

and

$$\text{Hom}((U'_1, \varphi_1), (U'_2, \varphi_2)) = \{f: U'_1 \rightarrow U'_2 \text{ commuting with everything}\},$$

that is, f such that the diagram

$$\begin{array}{ccc} U'_1 & \xrightarrow{f} & U'_2 \\ & \searrow & \swarrow \varphi_2 \\ & & X' \\ & \swarrow \varphi_1 & \\ & U & \end{array}$$

and f is G -equivariant.

We claim this is a group, and if G acts freely, then automorphisms of any point are trivial, and the groupoid is just a set.

Example 14. For $g \geq 0$ we want to define $\mathcal{M}_g$, the moduli stack of curves of genus g . For a scheme U , we want the groupoid $\mathcal{M}_g(U)$ to have objects $\pi: X \rightarrow U$ such that the geometric fibres of π are smooth projective curves of genus g . Morphisms will be isomorphisms of such things over U .

Example 15. Let k be a field and X be a smooth projective connected curve over k . Let $n \geq 1$ and $\text{Bun}_n(X)$ be the moduli stack of rank n vector bundles on X . For U a scheme over k , $\text{Bun}_n(X)(U)$ has objects the rank n vector bundles on $X \times U$ and morphisms given by isomorphisms of these last.

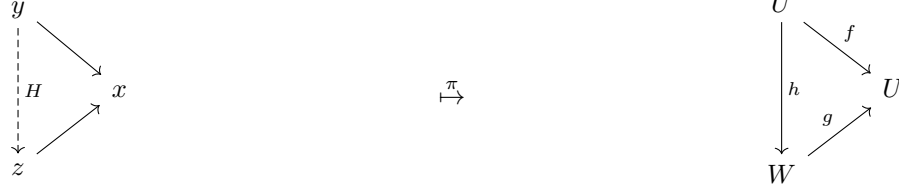
This will turn out to be a smooth algebraic stack of dimension $(g-1)n^2$, where g is the genus of X . Note that $n^2 = \dim_k \text{GL}_n(k)$. When $g = 0$, this dimension is negative. In the setting of stacks, this is permitted.

Now we will define stacks.

3.2.1 Categories fibred in groupoids

Definition 27. Fix a category T . A *category fibred in groupoids over T* is a category $\mathcal{X}$ and a functor $\pi: \mathcal{X} \rightarrow T$ such that

1. For all morphisms $f: V \rightarrow U$ in T , for all $x \in \mathcal{X}$ such that $\pi(x) = U$ and y such that $\pi(y) = V$, there exists a unique morphism $F: y \rightarrow x$ such that $\pi(F) = f$.
2. When there is $h: V \rightarrow U$ such that the diagram on the right commutes, then there is a unique H which makes the diagram on the left commute.



Why is this called a category fibred in groupoids? Take $u \in \text{Obj}(T)$ and consider the *fibre category* $\mathcal{X}(u)$ with $\text{Obj}(\mathcal{X})(u) = \{x \in \mathcal{X} \mid \pi(x) = u\}$ and $\text{Hom}_{\mathcal{X}(u)}(x, y) = \{F: x \rightarrow y \mid \pi(F) = \text{id}_u\}$. Note that the condition on the images of morphisms is a quotient-like condition.

Lemma 5. *For all $u \in \text{Obj}(T)$, $\mathcal{X}(u)$ is a groupoid.*

Proof. Apply lifting of triangles to the situation



to get H such that $F \circ H = \text{id}_x$. Further, by the same procedure, H itself has a right inverse $\tilde{F}$, and $FH\tilde{F} = F$ and on the other hand is equal to $\tilde{F}$. Therefore $F = \tilde{F}$ and $H = F^{-1}$. $\square$

Remark 7. For $f: U \rightarrow V$ and for all $x \in \mathcal{X}(V)$, there exists a pullback $y = f^*x \in \mathcal{X}(U)$.

Lemma 6. *f^*x is unique up to isomorphism.*

Proof.

todo

$\square$

Remark 8. A functor $\mathcal{X} \rightarrow T$ with the above properties is equivalent to the data of $\mathcal{X}(U)$ for all U and pullback functors.

Example 16. Let $T = \mathbf{Top}$ and G be a topological group. Consider $BG = \mathcal{X}$, the *classifying stack* of G . By definition $\text{Obj}(\mathcal{X})(U) = \{(U, P) \mid U \in \mathbf{Top} \text{ } P \text{ a principal } G \text{ bundle on } U\}$. That is, P has a G -action locally isomorphic to $U \times G$. Morphisms are diagrams

$$\begin{array}{ccc} P & \longrightarrow & P' \\ \downarrow & & \downarrow \\ U & \longrightarrow & U' \end{array}$$

Exercise 6. *Show that $\mathcal{X}$ satisfies the axioms.*

Every fibre category is the category of principal G -bundles over U . A similar definition works for $T = \mathbf{Sch}$, but one needs to carefully define principal G -bundles *i.e.* in what topology the bundle is to be locally trivial. If G is a linear algebraic group, a good choice is the étale topology.

3.2.2 Aside on 2-categories

Fix T . Categories fibred in groupoids over T form a 2-category. That is, there are objects, morphisms, and morphisms between morphisms. In a 2-category, $\mathcal{C} \text{ Hom}_{\mathcal{C}}(x, y)$ is not a set, but a category. We will avoid the rigorous definition here, because the naive (strict) definition of 2-category is too strict. For example, give objects x, y, z , composition is a functor

$$\text{Hom}(x, y) \times \text{Hom}(y, z) \rightarrow \text{Hom}(x, z).$$

Associativity in a 1-category is an equality: given objects x_1, x_2, x_3, x_4 , the two compositions are equal in the set $\text{Hom}(x_1, x_4)$. In a category, it is not a good idea to say that two objects are equal; one should instead say that $x_1 \simeq x_4$. An abstract isomorphism is however too weak. Therefore now associativity is additional data (a specified isomorphism) that should satisfy additional axioms.

Exercise 7. Let $T = \mathbf{Top}$ and let G_1, G_2 be topological groups. Describe the category of morphisms $BG_1 \rightarrow BG_2$ in terms of G_1, G_2 .

We want now to take $T = \mathbf{Sch}$ or $T = \mathbf{Sch}_S$ for a fixed scheme S . Our algebraic stacks will be categories fibred in groupoids over T , with additional properties. The definition will happen in two steps.

3.2.3 Pre-stacks and stacks over S

Definition 28. Let $\mathcal{X}$ be a category $\mathcal{X}$ fibred in groupoids over $\mathbf{Sch}_S$. Consider the following two possible conditions on $\mathcal{X}$:

1. for all $U \in \mathbf{Sch}_S$ and for all $x, y \in \mathcal{X}(U)$ the presheaf

$$\text{Isom}(x, y): \mathbf{Sch}_S^{\text{op}} \rightarrow \mathbf{Set}$$

defined by

$$(V \rightarrow U) \mapsto \text{Hom}_{\mathcal{X}(V)}(x_V, x_V),$$

where x_V, x_Y are pullbacks, is a sheaf in the étale topology. These pullbacks can also be written $x|_V$.

2. For all coverings $(V_i \rightarrow U)_i$ in $\mathbf{Sch}_S$ in the étale topology every descent datum $(x_i, f_{ij})_j$ relative to the above covering is effective.

We say that $\mathcal{X}$ is a *pre-stack over S* (also called an *S -groupoid*) if condition 1 holds. We say that $\mathcal{X}$ is a *stack over S* if conditions 1 and 2 both hold.

Here $x_i \in \mathcal{X}(V_i)$ and $f_{ji}: x_i|_{V_{ji}} \xrightarrow{\sim} x_j|_{V_{ji}}$, where $V_{ij} = V_i \times_U V_j$, satisfy the cocycle condition

$$f_{ki}|_{V_{kji}} = (f_{kj}|_{V_{kji}}) \circ (f_{ji}|_{V_{kji}}).$$

The first condition is a sheaf condition for morphisms, and the second, a sheaf condition for objects. Indeed, here *effective* means given the descent data, there is $x \in \mathcal{X}(U)$ such that $x_i = x|_{V_i}$ and all morphisms are the natural ones. Condition 1 and the fact that $\text{Isom}(x, y)$ is a sheaf implies that this x is unique up to canonical isomorphism.

Exercise 8. For all categories fibred in groupoids $\mathcal{X} \rightarrow T$ and for all $f: V \rightarrow U$ in T , we have a well-defined functor $f^*: \mathcal{X}(U) \rightarrow \mathcal{X}(V)$. (This requires both axioms.)

Remark 9. There exists a notion of *stackification* of a prestack. It follows the expected recipe, namely that the objects are themselves descent data. The nontrivial step in the construction is to show that the glueing condition for morphisms (condition 1) isn't destroyed in the process. The stackification construction is deserving of its name, as it provides a functor $i: \mathcal{X} \rightarrow \tilde{\mathcal{X}}$ from a prestack $\mathcal{X}$ to its stackification $\tilde{\mathcal{X}}$ which is left adjoint to the forgetful functor.

Example 17. Any scheme or (algebraic) space defines a stack. To begin, let T be a category and $F: T^{\text{op}} \rightarrow \mathbf{Set}$ be a functor. Then define

$$\text{Obj}(\mathcal{X}) = \{(u, s) \mid u \in T, s \in F(U)\}$$

and

$$\text{Hom}_{\mathcal{X}}((U, s), (V, t)) = \{f: U \rightarrow V \mid F(f)(t) = s\}.$$

finish pullbacks

Any space over S defines a category fibred in groupoids over S . In fact, it's a stack. Any functor $\mathbf{Sch}_S^{\text{op}} \rightarrow \mathbf{Set}$ defines a prestack.

3.3 Algebraic stacks

Our next task is to define algebraic stacks.

3.3.1 The diagonal morphism and representability

Definition 29 (Fibred product of groupoids over T). Given categories $\mathcal{X}'$, $\mathcal{X}''$ and $\mathcal{Y}$ fibred in groupoids over T and a diagram

$$\begin{array}{ccc} \mathcal{X}' & & \mathcal{X}'' \\ & \searrow F' & \swarrow F'' \\ & \mathcal{Y} & \end{array}$$

define $\mathcal{X} = \mathcal{X}' \times_{\mathcal{Y}} \mathcal{X}''$ by

$$\text{Obj}(\mathcal{X})(U) = \left\{ (x', x'', g) \mid x' \in \text{Obj}(\mathcal{X}')'(U), x'' \in \text{Obj}(\mathcal{X}'')''(U) \text{ } g: F'(x') \xrightarrow{\sim} F''(x'') \text{ in } \mathcal{Y}(U) \right\}$$

This groupoid fibred over T is naturally equipped with a functor $F: \mathcal{X} \rightarrow \mathcal{Y}$.

Exercise 9. Define the morphisms.

todo

Definition 30. We define the *diagonal morphism* $\Delta_{\mathcal{X}/\mathcal{Y}}$ to be

$$\Delta_{\mathcal{X}/\mathcal{Y}}: \mathcal{X} \rightarrow \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}.$$

It is a 1-morphism.

Proposition 4. 1. For all U , $\Delta_{\mathcal{X}/\mathcal{Y}}(U)$ is a faithful functor;

2. $\Delta_{\mathcal{X}/\mathcal{Y}}(U)$ is fully-faithful if $F(U): \mathcal{X}(U) \rightarrow \mathcal{Y}(U)$ is faithful;

3. The diagonal is an equivalence if $F(U)$ is fully-faithful.

Proof. Tautology. □

Definition 31. Given a morphism $F: \mathcal{X} \rightarrow \mathcal{Y}$ in categories fibred in groupoids, we say F is a *monomorphism* if $F(U)$ is fully-faithful for all U . (This is equivalent to requiring $F(U)$ be an equivalence for all U .)

Exercise 10. Recall we defined $\mathcal{X} = BG$ a category fibred in groupoids over $\mathbf{Top}$ for every topological group G . Let $G_1 \rightarrow G_2$ be a morphism of topological groups. We claim that we get a morphism $BG_1 \rightarrow BG_2$. Find out when this map is a monomorphism. Note that it does not suffice for the morphism of groups to be injective. For example $pt \rightarrow pt/G$ is not a monomorphism. On the LHS, over a point, the only morphisms are $\text{id}: pt \rightarrow pt$, but on the right-hand side, they are $\text{Hom}(pt, pt) = G$. Therefore $F(pt)$ cannot be fully-faithful.

Recall that a morphism of sheaves on a topological space is surjective if it is locally surjective on sections.

Definition 32. A morphism $F: Y \rightarrow \mathcal{X}$ is an *epimorphism* of stacks if for all U and every $x \in \mathcal{X}(U)$ there is a covering $U' \rightarrow U$ and $y' \in Y(U')$ such that $F(U')(y') \simeq x|_{U'}$ in $\mathcal{X}(U')$.

Lemma 7. A morphism of stacks is an isomorphism if it is a monomorphism and an epimorphism.

Proof. Exercise (not a tautology). $\square$

It makes sense to say that a stack $\mathcal{X}$ is representable by an algebraic space X .

Lemma 8. Given a stack $\mathcal{X}$ and a scheme U , we have

$$\mathcal{X}(U) \simeq \text{Hom}_{\mathbf{Stk}}(U, \mathcal{X}).$$

Proof. For all S -schemes V , given an object $x \in \mathcal{X}(U)$, we define

$$U(V) = \text{Hom}_{\mathbf{Sch}_S}(V, U) \rightarrow \mathcal{X}(V)$$

by

$$(f: V \rightarrow U) \mapsto f^*x \in \text{Obj}(\mathcal{X})(V).$$

This gives a map $\mathcal{X}(U) \rightarrow \text{Hom}_{\mathbf{Stk}}(U, \mathcal{X})$. To go the other direction, take $V = U$, and $f = \text{id}_U$. Apply the above to get an object of $\mathcal{X}(U)$. $\square$

Definition 33. We say that $\mathcal{X}(U)$ are the U -points of $\mathcal{X}$.

Definition 34. A 1-morphism $\mathcal{X} \rightarrow \mathcal{Y}$ is *representable* (resp. *schematic*) if for all schemes U and for all $y \in \mathcal{Y}(U)$, upon viewing y as a morphism $U \rightarrow \mathcal{Y}$, the fibre product $U \times_{\mathcal{Y}} \mathcal{X}$ is an algebraic space (resp. a scheme).

Example 18. Let $S = \text{Spec } k$ and G_1, G_2 be algebraic groups over k . Suppose we have $G_1 \rightarrow G_2$. Denote pt/G_i the classifying stack. When is $pt/G_1 \rightarrow pt/G_2$ representable?

First suppose that G_1 is trivial. Then $pt \rightarrow pt/G$ is representable.

A map $U \rightarrow pt/G$ is by definition a G -torsor $\tilde{U}$ over the scheme U (that is $\tilde{U}$ is a G -scheme etale locally isomorphic to $U \times G$). We claim that $\tilde{U}$ is the fibre product

$$\begin{array}{ccc} \tilde{U} \simeq pt \times_{pt/G} U & \longrightarrow & U \\ \downarrow & & \downarrow \\ pt & \longrightarrow & pt/G. \end{array}$$

Hence the fibres are $\tilde{U}$, a scheme. Note that the fibres horizontally are all G .

Now suppose that G_2 is trivial. Then the fibres of $pt/G \rightarrow pt$ are pt/G , which is not a scheme or an algebraic space.

The ultimate answer to the question is that the morphism is representable when G_1 is a closed subgroup of G_2 . Automorphisms of points have to do with the kernel of $G_1 \rightarrow G_2$, and our approximate slogan is that “representability $\iff$ pts in the fibre do not have automorphisms.”

Exercise 11. Above, prove that $\tilde{U}$ really is the fibred product.

Definition 35. A morphism Hence the fibres are $\tilde{U}$, a scheme. Note that the fibres horizontally are all G . is *representable by* (an open immersion, closed immersion ...) if it is schematic and for all schemes U , and $y \in \mathcal{Y}(U)$, $y: U \rightarrow \mathcal{Y}$, the morphism $\mathcal{X} \times_{\mathcal{Y}} U \rightarrow U$ of schemes is a (closed immersion, open immersion, ...).

3.3.2 Algebraic and Deligne-Mumford stacks

Definition 36. A stack $\mathcal{X}$ over S is *algebraic* (or an *Artin stack*) if

1. The diagonal morphism $\Delta: \mathcal{X} \rightarrow \mathcal{X} \times_S \mathcal{X}$ is representable, separated, and quasicompact
2. There exists an algebraic space X and $p: X \rightarrow \mathcal{X}$ that is representable, surjective, and smooth. We call X a *presentation of $\mathcal{X}$* .

Remark 10. Because every algebraic space has an étale cover by a scheme, we can always take the presentation of $\mathcal{X}$ to be a scheme.

The above definition means informally that an algebraic stack is a stack with an analogue of quasiseparatedness, that can also be nicely covered by a scheme.

Definition 37. A *Deligne-Mumford stack* is an algebraic stack $\mathcal{X}$ with a presentation $X \rightarrow \mathcal{X}$ that is representable, surjective, and étale.

The condition on the diagonal means explicitly that for all $x, y \in \mathcal{X}(U)$, $\text{Isom}(x, y)$ is a quasicompact and separated algebraic space, i.e. a sufficiently nice group scheme or group algebraic space when $x = y$.